



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
SEM 1 2025/2026
SECR3553 ARTIFICIAL INTELLIGENCE
(KEPINTARAN BUATAN)
SECTION 03

ASSIGNMENT 3:
PARKING DETECTION APPLICATION
THEME:
SMART CITY

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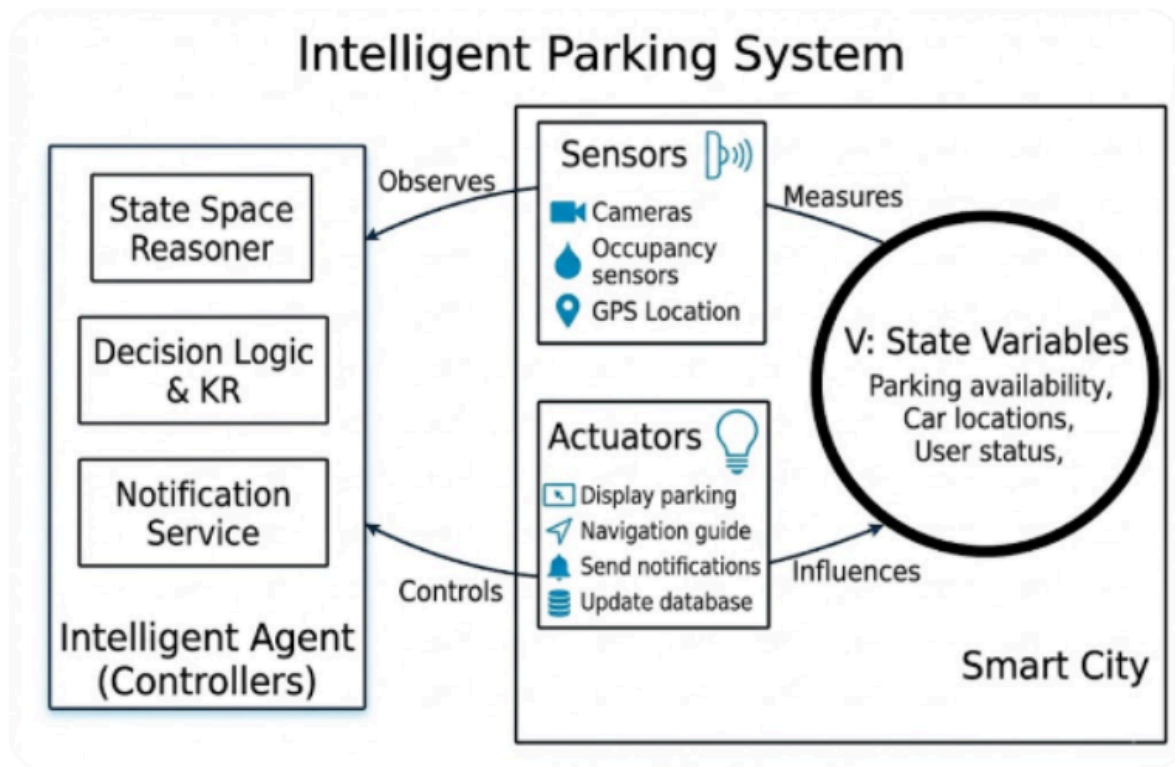
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1.0 PEAS Model

Performance measure	- Accuracy of parking space detection - Speed of real-time database updates - Reduction in parking search time - Timely notifications when parking is full
Environment	- Cities or campuses parking areas - Parking spaces - Vehicles entering and leaving - Driver using the application - Cameras, sensors and network infrastructure
Actuators/Effectors	- Updating the parking database - Displaying available parking spaces on the application - Providing navigation guidance - Sending notifications to drivers when parking is full
Sensors	- Surveillance cameras - Parking occupancy sensors - GPS location data from drivers - User requests from the application - Database status logs

2.0 Relations Between Each Property



Sensors collect real-time data from the parking environment and send it to the intelligent agent. In order to make decisions, the agent uses state-space reasoning and knowledge representation. After that, the actuators carry out tasks like delivering notifications, directing drivers and updating the database. A perception-decision-action loop is created when the environment is constantly changing and giving the sensors feedback.

3.0 Explanation In Proof Of Concept (POC)

Performance:

Environment:

Actuators/Effectors:

Actuators are represented through the actions performed by the system after analyzing parking availability data. When a parking space status changes, the system updates the parking database to maintain real-time accuracy and ensure consistency across all system components. The application then displays the current availability of parking spaces to users through a visual interface, allowing drivers to easily identify which parking slots are occupied or available. When a driver requests parking assistance, the system generates navigation guidance by identifying the nearest available parking space based on the driver's location and providing directional instructions to guide the user efficiently to the selected slot. Additionally, if all parking spaces are occupied, the system triggers a notification to inform users that the parking area is full, enabling them to avoid unnecessary entry and consider alternative parking locations. These actuators demonstrate how the intelligent agent responds to environmental inputs by updating information, guiding users to parking locations, and issuing alerts.

Sensors:

Sensors are represented as the input mechanisms that allow the intelligent agent to perceive its environment and receive user interactions. In the intelligent parking system, the sensors include surveillance cameras, parking occupancy sensors, driver location input (GPS), user interaction inputs from the application interface and database status logs that record changes in parking space occupancy. Parking space occupancy is captured through surveillance cameras or parking occupancy sensors, where each parking slot's status is identified as either occupied or vacant. The system also receives the driver's location information, which is used to determine the nearest available parking space. In addition, user actions, such as requesting parking availability or refreshing the system, serve as additional sensor inputs that trigger system responses. All detected changes and user interactions are recorded in the database to support continuous monitoring of the parking environment. These sensors enable the intelligent agent to gather relevant data, which forms the basis for intelligent reasoning, decision-making, and subsequent system actions.

4.0 Behaviours Of Agent To Achieve The System Goals